

Thermally evaporated perovskite/silicon tandems via formamidinium eutectic

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Solution processing remains the dominant route to high-performance perovskite/silicon tandems, but it remains challenging to simultaneously achieve industrial scalability and long-term reliability^{1–3}. Thermal evaporation is more industrially viable, yet it has not been successfully demonstrated for large-area perovskite/silicon tandems, largely due to the thermal degradation of formamidinium iodide (FAI) during high-temperature evaporation. Here we synthesize a formamidinium-based eutectic that lowers the effective evaporation temperature of FAI by an average of 36 °C—below its degradation threshold—thereby enabling stable FAI evaporation without thermal degradation. As a result, the evaporated perovskite films have enhanced crystallinity and atomic-scale compositional homogeneity. Sequentially evaporated perovskite/silicon tandems achieve a steady-state efficiency of 31.5% (1 cm²). Benefiting from the uniformity of evaporation, we further demonstrate a thermally evaporated large-area perovskite/silicon tandem on a commercial half-cut G12 wafer, delivering a steady-state efficiency of 30% (200 cm²). Scaling the device area from 1 cm² to 200 cm² incurs only a 3.99% relative efficiency loss, representing, to our knowledge, the lowest reported efficiency penalty for area scaling in perovskite-based tandems. The eutectic-based tandem retains 95% of its initial efficiency after 2,000 h of damp-heat ageing (85 °C and 85% relative humidity) and has negligible power loss after 2 months of real-world outdoor operation.

Integrating organic–inorganic metal halide perovskites atop crystalline silicon to form perovskite/silicon tandem solar cells offers a powerful route to boosting power conversion efficiency^{4–6}. However, state-of-the-art high-efficiency perovskite/silicon tandems predominantly rely on solution processing for perovskite deposition^{7–9}—a technique that remains challenging to reconcile with long-term reliability and industrial scalability^{10,11}.

Among the emerging fabrication approaches, vacuum thermal evaporation has attracted substantial attention owing to its intrinsic advantages (Supplementary Note 1)¹²; however, its application to the fabrication of large-area perovskite/silicon tandem devices has not yet been convincingly demonstrated or reported^{13,14}. A central bottleneck lies in the fundamental requirements of thermal evaporation. Perovskite film formation requires heating of the precursor materials—namely, inorganic PbX₂ and organic FAX (X = Cl, Br, I)—to elevated

temperatures under high vacuum to release molecular species for deposition. Organic precursors such as formamidinium iodide (FAI) are prone to thermal degradation during this high-temperature evaporation process, which is widely regarded as one of the most critical challenges in evaporated perovskite film fabrication (Supplementary Note 2)^{15–17}.

Two primary routes have been explored for thermally evaporated perovskite films^{18–20}: first, the one-step co-evaporation method, in which all precursors are evaporated simultaneously^{21–24}; and, second, the sequential two-step method, in which the inorganic lead halide layer is deposited first, followed by the organic layer^{25,26}. The two-step sequential method includes thermal evaporation combined with near-space sublimation, in line with recent works by Ballif et al.²⁷ and Bolink et al.²⁸, where a fully vacuum-processed perovskite/silicon tandem solar cell was realised with an efficiency of 24.3%. Although

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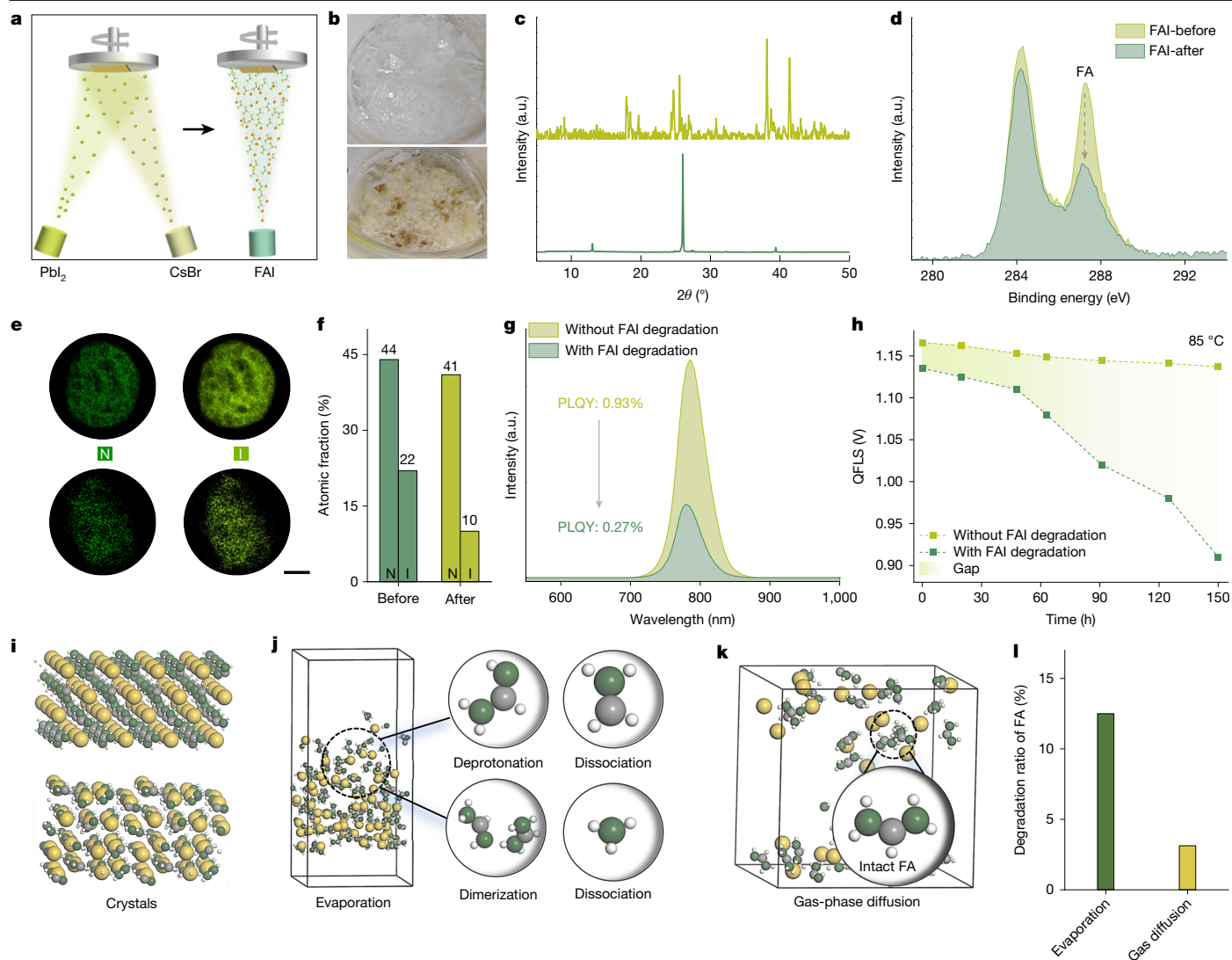


Fig. 1 | Thermal degradation of FAI. **a**, Schematic diagram of two-step sequential thermal evaporation for preparing perovskite films. **b**, Photographs of FAI raw materials before (top) and after (bottom) thermal evaporation in a quartz crucible. **c**, X-ray diffraction patterns of FAI before (bottom) and after (top) thermal evaporation. **d**, Carbon 1s XPS spectra of FAI before and after thermal evaporation. **e**, Transmission electron microscopy energy dispersive spectrometer mapping of FAI before (top) and after (bottom) thermal evaporation. Scale bar, 1 μm . **f**, The atomic ratio of nitrogen to iodine of FAI before and after thermal evaporation. **g**, Photoluminescence spectra

and PLQY of perovskite films with and without FAI degradation. **h**, The quasi-Fermi level splitting (QFLS) of perovskite films with and without FAI degradation recorded over time during ageing at 85 °C. **i**, (010) (top) and (111) (bottom) surfaces as the initial states of ML–MD simulations. H, white; C, grey; N, green; I, yellow. **j**, Representative image of the evaporation process during 1 ns ML–MD simulation, where the observed degradation products are highlighted. **k**, Representative image of the gas diffusion process after evaporation. **l**, Degradation ratios of FA molecules during the evaporation process or gas diffusion process.

the sequential two-step approach offers advantages such as higher deposition rates²⁹, the subsequent organic evaporation step requires higher temperatures, thereby exacerbating the degradation of organic precursors³⁰. This intrinsic instability of FAI has severely constrained the development and adoption of sequential thermal evaporation for perovskite/silicon tandem devices^{13,31,32}.

Herein we demonstrate severe high-temperature degradation of FAI during its thermal evaporation for perovskite film fabrication, which concomitantly degrades the optoelectronic performance and stability of the resulting devices. To address this challenge, we synthesize a formamidinium eutectic salt (Eu) that reduces the melting point of FAI by more than 100 °C. Compared with pure FAI, evaporation of Eu lowers the effective temperature at which FAI molecules escape from the bulk source and deposit onto the substrate by an average of 36 °C—which is below the degradation threshold—thereby enabling stable thermal evaporation of FAI without degradation. As a result, we

realize perovskite/silicon tandem solar cells fabricated via sequential two-step thermal evaporation, achieving a steady-state power conversion efficiency (PCE) of 31.5% (1 cm^2). Owing to the exceptional large-area compositional uniformity afforded by Eu evaporation, we further scale the technology to commercial half-cut G12 wafers, delivering large-area thermally evaporated perovskite/silicon tandem with a steady-state PCE of 30.0% (200 cm^2). Scaling the device area from 1 to 200 cm^2 results in only a 3.99% relative efficiency loss—the smallest reported for perovskite-based tandem devices and comparable with that of established silicon photovoltaics.

Thermal degradation of FAI

The two-step thermal evaporation method—which has higher efficiency potential but is more challenging—was adopted to prepare perovskite films (Fig. 1a). The inorganic lead halide and organic FAI layers were

deposited sequentially (Supplementary Fig. 1), forming an α -phase perovskite during annealing, which has micrometre-sized grains and symmetrical photoluminescence peaks (Supplementary Fig. 2). We first investigated the evaporation rate and chamber vacuum degree of FAI (Supplementary Note 3), and found that the evaporation of FAI in second step requires higher rate^{15,33} but under lower vacuum^{18,34}; thus, a higher temperature is needed, as previously reported¹². We further optimized the device performance for subsequent reliable mechanism research (Supplementary Fig. 6).

After thermal evaporation, we first observed that the remaining FAI in the quartz crucible had turned yellow or even black compared with the white crystals before evaporation (Fig. 1b and Supplementary Fig. 7), in line with a past report³⁵, which inspired us to further investigate whether FAI had degraded during its evaporation process. X-ray diffraction (XRD) revealed the formation of new diffraction peaks in remaining FAI in crucible after thermal evaporation (Fig. 1c and Supplementary Note 4). High-resolution X-ray photoelectron spectroscopy (XPS) also indicates that the proportion of FAI components in the substance left in the crucible after thermal evaporation has decreased (Fig. 1d). The energy dispersive spectrometer results from transmission electron microscopy (TEM) indicate that the N/I ratio of FAI before thermal evaporation is 2:1 (Fig. 1e,f and Supplementary Fig. 9), which is consistent with the stoichiometric ratio of its molecular formula $\text{CH}_3\text{N}_2\text{I}$. However, the N:I ratio had increased to nearly 4:1 in the substance left in the crucible after evaporation.

To gain insights into atomic degradation pathway of FAI, we conducted machine-learning-driven molecular dynamics (ML–MD) simulations to realize the large-scale full-workflow (from crystal to gas) intelligent reproduction of the FAI thermal evaporation process (Supplementary Table 1). As shown in Fig. 1i, according to our surface energies calculation the initial states of ML–MD simulations were dominant crystal surfaces of FAI, (010) and (111) (Supplementary Table 2). The evaporation process could then be observed in the ML–MD simulations, where FAI in the gas-state started to be released from the crystal surfaces (Fig. 1j and Supplementary Fig. 10). The degradation of FA molecules via various pathways becomes a common phenomenon during this process^{36,37}. The representative degradation reactions could be observed including deprotonation, dimerization and dissociation (Fig. 1j). Moreover, these FA degradation products remain in the vapour phase, and they may be incorporated into the final perovskite film. Meanwhile, in the following gas-phase ML–MD simulations, we found if the FAI molecules could be released intact into the gas phase, they could remain stable during the gas phase diffusion process (Fig. 1k). We observed that only 3% of FAI molecules degrade in the gas phase diffusion process, whereas as much as 12% undergo degradation during its evaporation process (Fig. 1l). The ML–MD results indicate that the most degradation-prone regime for FAI occurs during its evaporation from the solid state into the vapour phase. This highlights the critical role of the solid-state packing of FAI. To ensure the complete release of intact FAI molecules, a less constrained FAI molecular environment (lower dissociation energy) is required.

We conducted time-of-flight secondary ion mass spectrometry tests (Supplementary Fig. 11), and the degradation product of FAI was detected in the perovskite film obtained by thermal evaporation^{36,37}. The vapour phase degradation products were further confirmed by gas chromatography–mass spectrometry (Supplementary Fig. 31). These results firmly confirm that FAI has actually been degraded during its thermal evaporation. Next, we investigated whether the thermal degradation of FAI will compromise the quality of obtained perovskite films. As shown in Supplementary Fig. 12, we first prepared the same batch of halide lead layer through thermal evaporation, we then spin-coated the residual FAI in the crucible (representing the degraded samples) and the fresh FAI (representing the undegraded samples) to prepare the perovskite films. We found that the perovskite film prepared by residual FAI from the crucible had significantly lower

photoluminescence intensity and photoluminescence quantum yield (PLQY; Fig. 1g). We also tracked their thermal stability under 85 °C ageing. We found that the stability of the perovskite film prepared using degraded FAI was markedly lower (Fig. 1h), the difference in QFLS values between them gradually increases. Overall, we found that FAI had degraded during its thermal evaporation, which would weaken the photoelectric performance and stability of the final thermally evaporated perovskite films. This might thus be a fundamental factor that is severely limiting the performance of thermal evaporation prepared perovskite solar cells.

Formamidinium eutectic

A viable solution to avoid thermal degradation of FAI is to lower its evaporation temperature (Supplementary Note 5). Inspired by the concept of organic eutectic, which is synthesized by hydrogen bond donor and acceptor materials, the formation of the hydrogen bond network enables the melting point of the eutectic phase to be dramatically reduced (Supplementary Note 6). We therefore attempted to use choline chloride (ChCl, the most classic hydrogen bond acceptor for a eutectic) to synthesize the formamidinium eutectic (Supplementary Fig. 16 and Supplementary Note 7). There are two criteria to determine whether it is a eutectic phase: first, whether a new single phase is formed rather than a physical mixture; and, second, whether the melting point is effectively lowered. First, proton nuclear magnetic resonance (¹H NMR) test results of both the liquid and solid states demonstrate that ChCl was included in the synthesized Eu (Supplementary Figs. 19 and 20). The XRD results show that the obtained Eu does not have any overlapping diffraction peaks with the original FAI, proving the synthesis of a new phase rather than a physical mixture (Supplementary Figs. 21 and 22). High-resolution TEM (HRTEM) lattice fringes and the corresponding fast-Fourier-transform show that Eu and FAI have markedly different diffraction spots (Fig. 2a and Supplementary Fig. 23), and that Eu has a larger interplanar spacing; these results firmly prove that the new phase was synthesized.

Hydrogen bonding is a prerequisite for the formation of an organic eutectic because the hydrogen bond network disrupts the long-range order of the crystal. For the ¹H NMR test of FAI, the resonance signal of the hydrogen atoms in $-\text{NH}_2$ was located at 8.75 ppm, whereas it broadened and shifted to 8.90 ppm in Eu (Fig. 2b). Such changes indicated the reduced electron shielding around the proton on $-\text{NH}_2$ (ref. 38). The Fourier-transform infrared spectroscopy results reveal that the stretching vibration frequency of the N–H bond in FAI shifted to a lower wave number from 3,352 to 3,286 cm^{-1} after forming Eu (Fig. 2c), manifesting the delocalization of the electronic cloud of the N–H bond³⁹. Furthermore, compared with pure ChCl, the XPS peak of Cl shifted towards a higher binding energy by 0.3 eV after forming Eu (Fig. 2d), which is indicative of decreased electron cloud density around the Cl atoms. The formation of a N–H $\cdots$ Cl hydrogen bonding network could be reasonably concluded on the basis of these results (Fig. 2e). This is because the formation of N–H $\cdots$ Cl hydrogen bonding induces mutual polarization of electron clouds towards the interatomic bonding region, with the total electron density strictly conserved. This spatial redistribution depletes local electron density around both the N–H proton and chloride anion. Furthermore, the hydroxyl group on ChCl also forms hydrogen bonds with the I on FAI by O–H $\cdots$ I (Supplementary Note 8), in turn forming hydrogen bonding networks (Fig. 2e). Density functional theory (DFT) calculations demonstrate possible types of molecular interactions in the Eu system (Supplementary Fig. 25a), where the interlaced hydrogen bonding network formed by FAI and ChCl has the highest interaction energies compared with the self-interaction of FAI or ChCl (Supplementary Fig. 25b), implying that the formation of Eu is energetically favoured. The DFT results further indicate that the Eu system has a markedly lower dissociation energy compared with pure FAI (Fig. 2f and Supplementary Fig. 26), suggesting

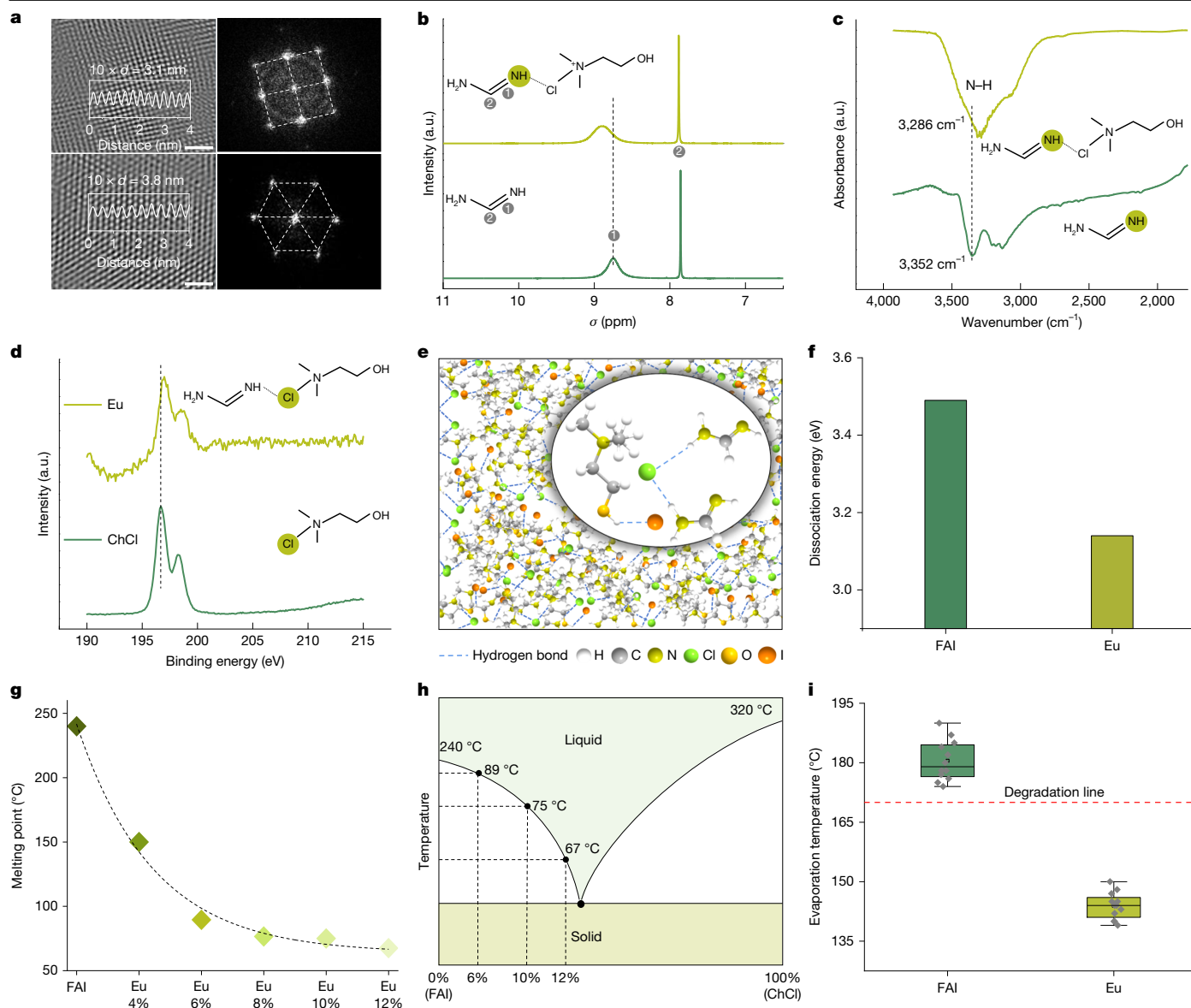


Fig. 2 | Synthesis of formamidinium eutectic. **a**, High-resolution TEM lattice fringes and corresponding fast Fourier transform diffraction spots of FAI (top) and Eu (bottom). Scale bars, 2 nm. **b**, ^1H NMR result of Eu (top) and FAI (bottom). **c**, Fourier transform infrared spectroscopy results of Eu (top) and FAI (bottom). **d**, X-ray photoelectron spectroscopy of Cl $2p$ in Eu (top) and ChCl (bottom). **e**, Schematic diagram of the hydrogen bond network in Eu. **f**, Dissociation

energies of Eu and FAI. **g**, Melting points of FAI and Eu with different ChCl contents, average values measured by DSC and heating experiments. **h**, Schematic phase diagram of binary phase of FAI and ChCl. **i**, Statistics of the evaporation temperature of Eu-4% and FAI, based on 24 independent experiments for perovskite film preparation. $P = 10^{-2}$ Pa; $R = 2 \text{ \AA s}^{-1}$.

that lower energy is required for thermal evaporating Eu, and thus lower temperature.

Experimentally, we heated a series of Eu with different ChCl contents under atmospheric pressure to observe the changes in states of matter. As shown in Supplementary Fig. 27, they are all solid powders at room temperature, but just by raising the temperature to 70°C , the Eu-12% sample begins to melt. It is noteworthy that the melting points of pure FAI and ChCl are 240°C and 302°C , respectively, which are much higher than 70°C . Raising the temperature further to 85°C caused the melting of Eu-6%, whereas Eu-12% has completely transformed into a liquid state. At just 100°C , we could even obtain a transparent formamidinium eutectic solution, which unambiguously confirms the effectively reduced melting point. The melting points of Eu were further measured through differential scanning calorimetry (DSC; Supplementary Fig. 28), which revealed that the synthesis of the eutectic salt dramatically reduced the melting point of FAI by more than 100°C

(Fig. 2g). The measured melting point allows us to outline the phase diagram of the binary phase of FAI and ChCl (Fig. 2h). The low molecular binding of FA in eutectic system arises from the combined effects of reconstructed ionic interactions, rearranged hydrogen bonding, disrupted long-range lattice order, and enhanced mixing entropy (Supplementary Note 9). We chose Eu-4% for subsequent thermal evaporation preparation of perovskite films and found that under real evaporation conditions (10^{-2} Pa), it had already turned into a molten state just at 120°C (Supplementary Fig. 29). We recorded the thermal evaporation temperature in the second step for the real preparation of perovskite films under the same vacuum degree (10^{-2} Pa) and evaporation rate ($2 \text{ \AA s}^{-1}$). Compared with the evaporation of FAI, the evaporation of Eu requires a dramatically lower temperature, with an average reduction of 36°C and a lowest temperature of only 139°C (Fig. 2i). Such a significant temperature drop may prevent the thermal degradation of FAI, thereby allowing us to obtain higher-quality perovskite films.

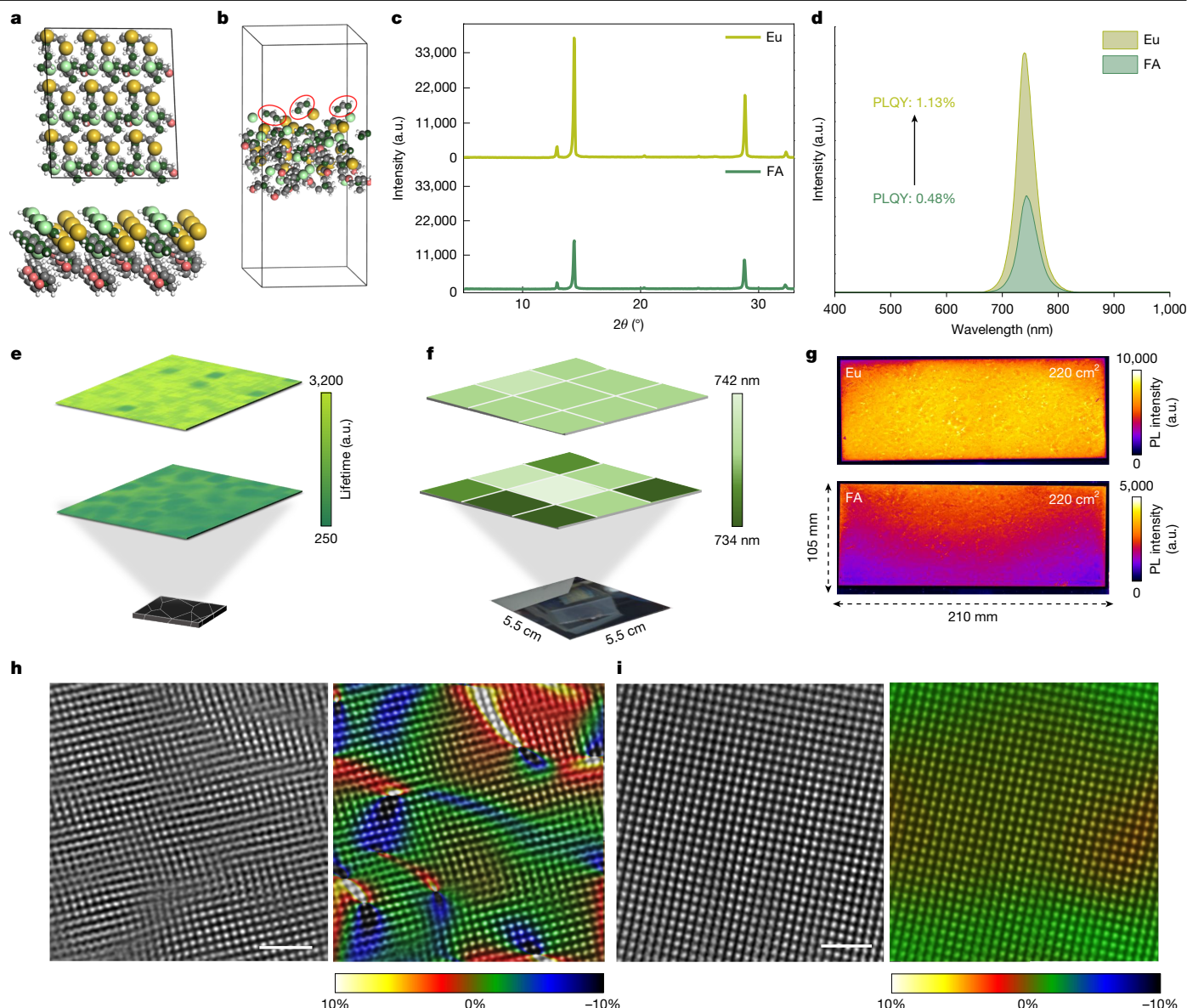


Fig. 3 | Characterizations of perovskite films prepared by thermal evaporation. **a**, Simulation of Eu evaporation. Top: top-down view of initial state of Eu for ML–MD simulation. Bottom: frontal view of the initial state of Eu for the ML–MD simulation. Carbon, grey; nitrogen, green; hydrogen, white; iodine, orange; chlorine, cyan; oxygen, red. **b**, Snapshot of ML–MD simulation, showing evaporated FA in the red circles. To simplify the model reasonably, we only constructed FAI-based Eu without introducing FABr. **c**, X-ray diffraction comparison of FA- and Eu-based perovskite films. **d**, Photoluminescence intensity and PLQY comparison of FA- and Eu-based perovskite films.

Quality of thermally evaporated perovskite films

To assess the impact of suppressing FAI degradation on the perovskite film, we replaced the conventional FAI (labelled FA throughout the paper) with lower-evaporation-temperature Eu. Although Eu is in a molten state during its thermal evaporation process, it can be controllably evaporated with a stable evaporation rate (Supplementary Fig. 30a). We first conducted ML–MD simulations to reveal the evaporation of Eu (Fig. 3a); the release of intact FA and FAI molecules is observed (Fig. 3b). We further elaborate on the physical picture of Eu thermal evaporation (Supplementary Note 10), in which Eu hardly produces distinct gaseous species to affect the lattice assembly of perovskite or the crystallization kinetics of the films (Supplementary Notes 11 and 12).

e, Time-resolved photoluminescence mapping of Eu-based (top) and FAI-based (middle) perovskite films ($20\ \mu\text{m} \times 20\ \mu\text{m}$). **f**, Photoluminescence emission peak wavelengths of Eu-based (top) and FA-based (middle) perovskite films ($5.5\ \text{cm} \times 5.5\ \text{cm}$). **g**, Large-area photoluminescence intensity mapping. **h**, High-resolution TEM image (left) for micro-strain measurement (right) of FA-based perovskite film. **i**, High-resolution TEM image (left) for micro-strain measurement (right) of Eu-based perovskite film. The colour bars indicate the percentage of micro-strain. Scale bars, $2\ \text{nm}$ (**h**, **i**).

Unlike the emergence of new diffraction peaks in residual FAI after thermal evaporation (Fig. 1c), the XRD diffraction peaks of Eu remain nearly identical after thermal evaporation (Supplementary Fig. 30b), indicating effective avoidance of thermal degradation. Thermal evaporation of Eu may not affect the bandgap and composition of the resulting perovskite (Supplementary Fig. 35); Eu leads to slightly larger grain (Supplementary Figs. 36 and 37) and substantially enhanced crystallinity with higher XRD intensity (Fig. 3c). The perovskite films prepared by evaporating Eu have twofold higher photoluminescence intensity and PLQY than those prepared by evaporating conventional FAI (Fig. 3d and Supplementary Fig. 38). Time-resolved photoluminescence (TRPL) mapping also indicates that thermal evaporation of Eu substantially enhances the carrier lifetime of perovskite film (Fig. 3e), confirming

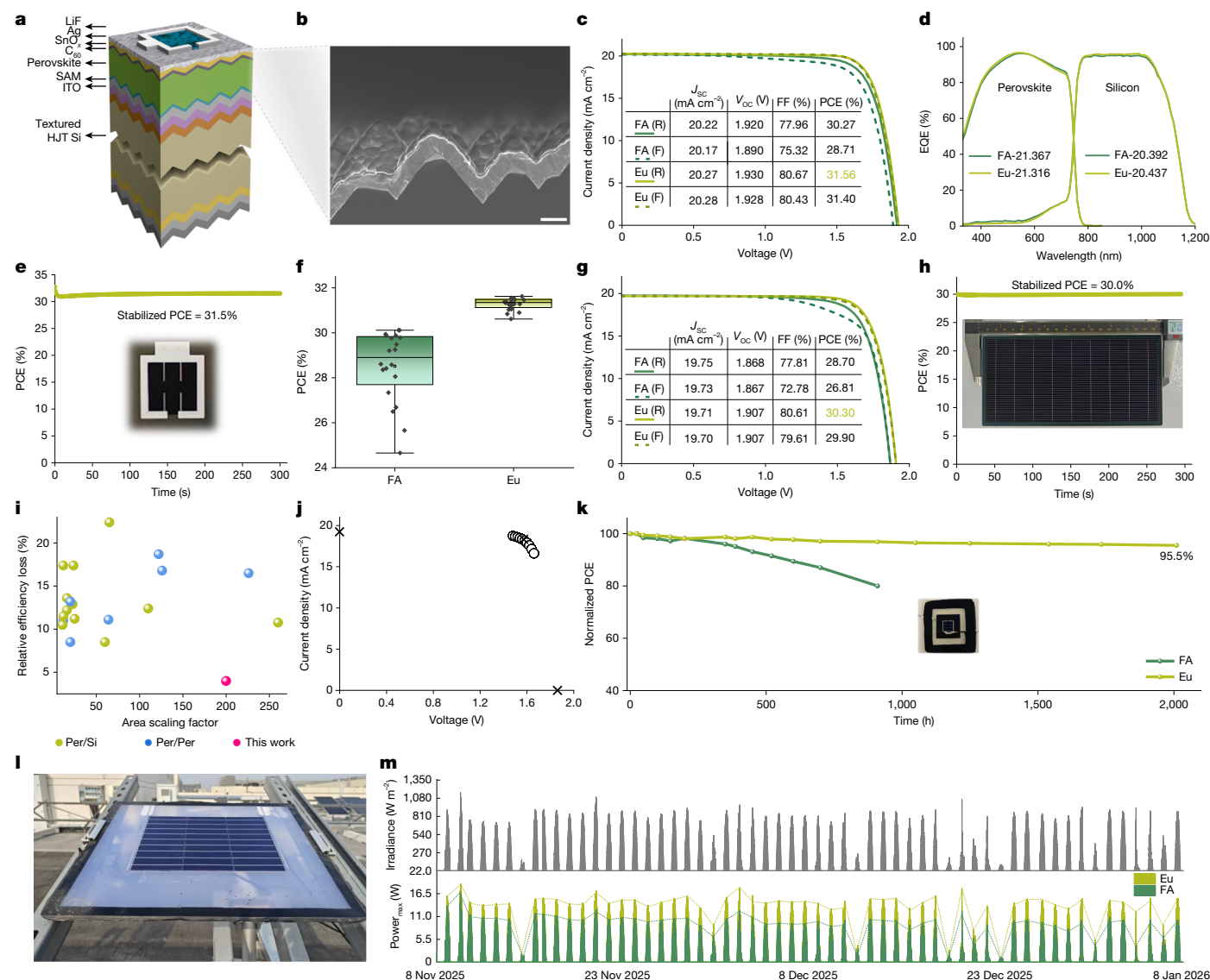


Fig. 4 | Photovoltaic performances. **a**, Schematic diagram of the tandem device. **b**, Cross-sectional SEM image of the tandem device. Scale bar, 1 μm. **c**, J - V curves of small-area tandem devices (1 cm²). **d**, External quantum efficiency spectra of small-area tandem devices. **e**, Stabilized power output of small-area tandem devices. The inset is a photograph of the device. **f**, The statistical distribution of PCE of FA- and Eu-based small-area tandem devices. **g**, J - V curves of large-area tandem devices (200 cm²). **h**, Stabilized power output of large-area tandem devices (200 cm²). The inset is a photograph of the device. **i**, Comparison of efficiency loss caused by area scaling (data were

extracted from Supplementary Tables 6 and 7). **j**, Certification results of large-area tandem devices (147.5 cm²) at the NREL, USA. The device duration was 734 s and the PCE was (29 ± 1.02)%. The crosses from left to right represent the J_{sc} , maximum power and V_{oc} points, respectively. **k**, Damp heat test of FA- and Eu-based tandem devices under 85 °C and 85% relative humidity. **l**, Photograph of the perovskite/silicon tandem devices under outdoor operation. **m**, Outdoor irradiance record and power output of the perovskite/silicon tandem module, with initial efficiencies of the Eu and FA devices of 26.5% and 24.0%, respectively.

that the avoided FAI degradation results in a lower defect density in thermally evaporated perovskite films (Supplementary Note 13). We confirmed that the improved perovskite film is mainly due to the avoidance of FAI thermal degradation (Supplementary Note 14)⁴⁰. We also observed that Eu resulted in better buried interfaces (Supplementary Note 15)^{41,42}.

Kelvin probe force microscopy mapping reveals that FA-based perovskite films have a relatively inhomogeneous surface potential (Supplementary Fig. 44). High-resolution TEM revealed that there was pronounced micro-strain in FA-based perovskite film (Fig. 3h), but it was effectively suppressed by evaporating Eu (Fig. 3i). We measured nine photoluminescence peak positions of FA- and Eu-based perovskite films on 5.5 cm × 5.5 cm films (Fig. 3f). The FA-based perovskite film had poorer phase homogeneity, with a maximum difference of 8 nm in the position of the emission peak. By contrast, the Eu-based perovskite film

still maintained the homogeneous phase distribution. We further prepared 220 cm² large-area perovskite films (Fig. 3g); photoluminescence intensity mapping revealed that the thermal evaporation of Eu still demonstrated exceptional homogeneity. The inherent advantages of the two-step sequential thermal evaporation—together with single-source evaporation of halides and suppression of FA degradation—collectively contribute to large-area film homogeneity (Supplementary Note 16). Furthermore, the improved film quality has been re-confirmed at the device level (Supplementary Note 17)⁴³.

Eu for devices

We fabricated perovskite/silicon tandem solar cells via two-step sequential deposition on industrial-grade micrometre-scale heterojunction silicon bottom cells (Fig. 4a), achieving conformal perovskite

film coverage (Fig. 4b). The current density–voltage (J – V) curves show that the champion PCE of a FA-based tandem device is 30.27% (Fig. 4c), whereas it was enhanced to 31.56% in a Eu-based tandem device (Fig. 4c), which is higher than the reported efficiency for perovskite/silicon tandems fabricated through thermal evaporation methods (Supplementary Table 4). Moreover, Eu-based tandem devices have negligible hysteresis with a hysteresis factor of 0.5%. The integrated J_{sc} from external quantum efficiency (EQE) measurements of the FA- and Eu-based sub-cells are 21.36 mA cm^{−2} and 21.31 mA cm^{−2}, respectively (Fig. 4d), aligned with the J – V curve. The fully textured pyramid structure contributed to the high J_{sc} . After continuous operation under maximum power point tracking (MPPT) for 600 s, the PCE of the Eu-based tandem device stabilized at 31.5% (Fig. 4e). The photovoltaic parameters of 20 devices from the same batch were statistically analysed (Supplementary Fig. 50), which revealed that Eu-based devices have a higher average V_{oc} and fill factor. The Eu-based devices achieved an average PCE of 31.28%, much higher than 28.51% of FA-based devices (Fig. 4f).

Commercial manufacturing requires high-rate deposition to improve mass production efficiency^{35,44}. The Eu-based device offers a higher maximum thermal evaporation rate without significant degradation of device performance (Supplementary Note 18; the reusability of Eu is evaluated in Supplementary Fig. 55). The ultimate commercialization requires large-area perovskite solar cells, but due to the challenge of inhomogeneity in large-area perovskite films, scaling up the area of perovskite film always results in significant efficiency loss in final device, which is a critical barrier currently hindering viable commercialization pathways of perovskite photovoltaic technology. Taking advantage of exceptional homogeneity of evaporating Eu, we attempted to fabricate perovskite/silicon tandem solar cells on a commercial half-cut G12 wafer (200 cm²). The champion efficiency of the Eu-based large-area tandem device reached 30.3% (Fig. 4g), which is among the highest efficiency reported for large-area perovskite/silicon tandem solar cells so far (Supplementary Table 5). After 300 s of MPPT, the efficiency of large-area tandem device stabilized at 30% (Fig. 4h). More importantly, after the area was increased to 200 cm², the efficiency loss of our Eu-based tandem devices is only 3.99% (that is, $\left(\frac{31.56 - 30.3}{31.56} \times 100\right)\%$, which represents the smallest efficiency loss caused by scaling up the area of perovskite-based tandem devices (Fig. 4i and Supplementary Tables 6 and 7)^{9,45–47}. We sent a large-area tandem device to the National Renewable Energy Laboratory (NREL) in the US for efficiency certification, but due to the size limitation of their light sources, we had to cut the tandem device to 147.5 cm². A certified quasi-steady state PCE of (29 ± 1.02)% was obtained (Fig. 4j and Supplementary Figs. 56–58).

We found that the Eu-based device showed almost no performance degradation after being stored in nitrogen for over 3,600 h (Supplementary Fig. 59). The Eu-based device also had markedly superior stability under MPPT (Supplementary Fig. 60). To further evaluate the stability of the tandem devices under harsher conditions, they were aged under 85 °C and 85% relative humidity for 2,000 h. The Eu-based tandem device still maintained more than 95% of its initial efficiency, making it the most stable perovskite/silicon tandem device against damp-heat testing (Fig. 4k, Supplementary Fig. 61 and Supplementary Table 8). Due to the removal of the thermally unstable surface passivator, the initial efficiencies of the Eu and FA devices in the damp heat stability test were 28.8% and 27.1%, respectively. To assess the real power generation stability of tandem devices, our assembled tandem modules were operated outdoors at a ground-based test facility (Fig. 4l) under MPPT. The power generation was recorded every 30 s, with a total of about 170,000 data points collected. From the real-time measured module power (Fig. 4m), the Eu-based tandem module demonstrates no power generation loss after two months of outdoor operation, firmly confirming its exceptional stability (Supplementary Note 19).

Conclusion

We have demonstrated the thermal degradation of FAI during its evaporation through cross-validation of various characterization techniques. We then synthesized formamidinium eutectic salt to dramatically lower the melting point of FAI by over 100 °C, allowing us to achieve effective thermal evaporation at lower temperatures to avoid its thermal degradation. As a result, we report perovskite/silicon tandem solar cells manufactured by sequential two-step thermal evaporation with a steady-state PCE of 31.5%. Critically, by leveraging the exceptional large-area component homogeneity of Eu evaporation, a high steady-state PCE of 30% was maintained, even when the area of the perovskite/silicon tandem device was scaled to a commercial half-cut G12 wafer (200 cm²). This work represents a successful fabrication of large-area perovskite/silicon tandem solar cells via thermal evaporation, which had the smallest area-scaling efficiency loss factor, thus establishing a viable route toward industrial translation of tandem technology.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10970-1>.

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Methods

Materials

Caesium bromide (CsBr, 99.99%), lead chloride (PbCl₂, 99.99%) and lead iodide (PbI₂) were purchased from Alfa. Methylammonium chloride and ethane-1,2-diammonium iodide were purchased from Greatcell Solar. FAI was purchased from Advanced Election Technology. Choline chloride, urea, isopropyl alcohol, ethanol and anisole were purchased from Sigma-Aldrich. MeO-4PACz was purchased from Tokyo Chemical Industry. Silver was purchased from Kurt J. Lesker. C₆₀ was purchased from Lumtec.

Preparation of formamidinium eutectic salt

To synthesize the formamidinium eutectic salt for WBG perovskite, we dissolved different molar ratios of FAI (3,440 mg), FABr (3,750 mg) and CHCl in 12 ml of isopropyl alcohol; the solution was then heated and stirred at 75 °C for 30 min to ensure complete dissolution; it was then gradually dropped into 120 ml of anisole solution, and the white precipitate could be obtained immediately. The anisole was then thoroughly removed from the precipitate, and then the white precipitate was transferred to a vacuum drying oven at 65 °C to dry for 48 h. It should be noted that the synthesis process needs to be performed in a glovebox filled with nitrogen. Exposure of the eutectic salts to the atmosphere should be minimized as much as possible during the transfer of the wet eutectic salts to the vacuum drying oven.

Preparation of perovskite films by sequential thermal evaporation

In the first step of evaporation, the evaporation rates of PbI₂, CsBr and PbCl₂ were maintained at 1 Å per second (400 °C), 0.1 Å per second (490 °C) and 0.1 Å per second (415 °C), respectively. An approximately 500-nm-thick inorganic lead layer was deposited at a vacuum degree of 10⁻⁴ Pa and substrate rotation speed of 20 r.p.m. In the second step, 3,000 mg of formamidinium eutectic salt (Eu-4%) was loaded into a quartz crucible for thermal evaporation at a rate of approximately 2 Å per second, while methylamine chloride was simultaneously evaporated at a rate of 0.8 Å per second. Each thermal evaporation will consume approximately 500 mg of Eu. It should be noted that for organic evaporation, slow heating and adequate preheating are needed to stabilize the rate before turning on the sample shutter. An organic layer of about 500 nm was then deposited at a vacuum degree of 10⁻² Pa and substrate rotation speed of 20 r.p.m. The films were then transferred to the atmosphere and annealed at 150 °C for 15 min to obtain the perovskite films. For the preparation of the control sample, in the first step, PbBr₂ needs to be evaporated at a rate of 0.6 Å per second (380 °C) to ensure the same band gap as the perovskite obtained by evaporating eutectic salt in the second step. For the control sample in the second step, FAI, FABr and CHCl were evaporated from three separate sources.

Silicon heterojunction bottom-cell fabrication

The silicon bottom cells were fabricated using n-type Czochralski silicon wafers with a resistivity of 1–5 Ω cm and a thickness of approximately 250–300 μm. Initially, a double-sided symmetrical texturing process was then performed using a potassium hydroxide solution, resulting in random pyramids with a height of 1,000–1,500 nm on both surfaces. The wafers then underwent a standard RCA (Radio Corporation of America) cleaning procedure and were then loaded into a plasma-enhanced chemical vapour deposition cluster for the passivation and doped layers. Hydrogenated intrinsic amorphous silicon layers—each with a thickness of about 10 nm—were deposited on both the front and rear surfaces. This was followed by the deposition of a 20-nm-thick n-type hydrogenated nanocrystalline silicon layer and a 30-nm-thick p-type hydrogenated nanocrystalline silicon layer on the front and rear side of the wafer, respectively. A 150-nm-thick indium tin oxide (ITO) layer was sputtered onto the rear side through a shadow mask, defining

an active area of 1.2 × 1.2 cm² for small cells and 20 × 9.5 cm² for large cells. Silver rear electrodes were then screen-printed onto the ITO layer, also through the same mask pattern. Finally, a 10-nm-thick ITO layer, serving as the tunnel recombination layer, was deposited on the front side. For small cells, the completed silicon bottom cells were laser-cut into 2.5 × 2.5 cm² substrates for subsequent tandem cell fabrication.

Perovskite/silicon tandem cells fabrication

The textured Si sub-cell was first treated with ultraviolet ozone for 13 min and then immediately transferred to a glovebox filled with nitrogen. The MeO-4PACz (0.8 mg ml⁻¹ in ethanol) was spin-coated on the Si substrate at 3,000 r.p.m. for 30 s and annealed at 100 °C for 10 min. Ethanol was then used to dynamically spin coat and wash the surface of unanchored MeO-4PACz molecules to achieve a monolayer. The samples were then transferred to the thermal evaporation chamber for subsequent deposition of perovskite layers. After the preparation of the perovskite film, ethane-1,2-diammonium iodide—dissolved in IPA at a concentration of 0.5 mg ml⁻¹—was dynamically dropped onto the perovskite film to passivate surface defects, and then the film was annealed at 100 °C for 6 min. Then, 15 nm C₆₀ was deposited as an electron transport layer by thermal evaporation. After that, 20 nm SnO₂ was deposited by atomic-layer deposition. The samples were then transferred to a magnetron sputtering chamber to deposit 60 nm ITO as the top transparent electrode. Subsequently, a 500 nm Ag frame was deposited by thermal evaporation. Finally, 80 nm LiF was thermally evaporated to serve as an anti-reflection layer.

Material characterizations

The crystal structure and phase composition were determined by the XPert Pro X-ray diffractometer with Cu Kα1 radiation (λ = 0.154 nm). Confocal laser scanning microscopy (CLSM) characterization was performed on a custom-built Olympus FV3000 CLSM system with a ×40 objective lens. The surface and cross-sectional morphology of the perovskite films and devices were recorded using a scanning electron microscope (Quanta FEG 450) under an electron-beam acceleration voltage of 10-kV. The spatial distribution of the atomic force microscopy infrared spectroscopy readings was characterized by a nanoIR2 system (Anasys Instruments). Both height image and chemical image were obtained in the contact mode using a contact-mode atomic force microscopy probe for nanoIR2 in the ambient atmosphere. Time-of-flight secondary-ion mass spectrometry was performed under an energy of 30 keV and current of 0.43 pA. Raman spectroscopy was conducted on the Horiba LabRAM HR Evolution using a 532 nm laser and a laser power of 5 mW.

Transmission electron microscopy specimen fabrication and characterization: powder specimens of FAI- and Eu-based samples were prepared by dispersing the powders onto ultrathin amorphous carbon films inside of a nitrogen-filled glovebox to prevent degradation. For device characterization, cross-sectional samples of perovskite films and solar cells were prepared using a dual-beam focused ion beam-scanning electron microscope (Helios NanoLab 450S, Thermo Fisher Scientific). A protective multilayer capping was applied—consisting of evaporated amorphous carbon, sputtered Au and ion-deposited Pt—to preserve the perovskite structure during milling. Lamellae were extracted using Ga⁺ ions at 1–30 kV; these were then thinned and polished at 2 kV and 23 pA to minimize damage. High-resolution TEM and high-angle annular dark-field scanning TEM were conducted on Cs-corrected Spectra-300 microscopes operated at 200 or 300 kV, achieving sub-ångström resolution. To reduce beam-induced degradation, HRTEM images were recorded on a Gatan K3 direct-detection electron-counting camera under dose-fractionation mode, with an electron dose rate maintained at around 5 e⁻ Å⁻² s⁻¹ and a total cumulative dose below 100 e⁻ Å⁻². Elemental analysis was performed using energy-dispersive X-ray spectroscopy on a Super-EDS system integrated with the Spectra-300, equipped with dual windowless silicon drift detectors, which enabled high-sensitivity elemental mapping.

Computational details

Structural database was established as the training data of the machine learning interatomic potential (MLIP) for the FAI. We have designed various structures—including FAI species in the bulk, surface and gas phase, as well as possible degradation products of FAI—to guarantee the global sampling of potential energy surface of FAI (Supplementary Table 1). The MLIP was constructed and trained on the basis of 19,500 structures that have been calculated by first-principles. Large-scale ML–MD simulations were performed with the trained MLIP.

The construction of neural network potentials of FAI and Eu was performed using LASP (large-scale atomistic simulation with neural network potential) code⁴⁸. The training dataset was generated by potential energy surface sampling driven by stochastic surface walking⁴⁹ exploration based on DFT calculations. Two training datasets were designed and constructed for the evaporation process for FAI and Eu separately. In these datasets, we have constructed three main kinds of structures with different features. First, bulk structures of FAI (Eu) as well as FAI (Eu) crystals with different defects that contain degradation products of FA. Second, different crystal surface structures of FAI (Eu) with free possible FA molecules on the surface to simulate the process of evaporation (Supplementary Fig. 1). Third, gas phase structure with FA (Eu) and iodide ions distributed randomly in the lattice as well as possible degradation products. All of these structures were optimized using the global optimization method (stochastic surface walking) to generate the trajectory structures as the sampling of potential energy surface. For the 19,500 chosen structures for FAI and 14,000 chosen structures for Eu in the training dataset, we further conducted high-accuracy electronic self-consistent calculation to correct the energy, force and stress information for each structure. Hyperbolic tangent activation functions were then used for the hidden layers (network framework: 558-128-80-80-1 for FAI potentials and 1073-128-80-80-1) during neural network fitting, whereas a linear transformation is applied to the output layer of all networks. The limited-memory Broyden–Fletcher–Goldfarb–Shanno method was used to minimize the loss function to match DFT energy, force and stress.

First-principles calculations were performed using the plane-wave pseudopotential method based on the Perdew–Burke–Ernzerhof exchange–correlation functional⁵⁰, which was implemented with the Vienna ab initio simulation package (v.5.4.4)⁵¹. The valence wave functions were expanded by plane wave with a cutoff energy of 400 eV, whereas all of the structures were optimized until the force on each atom was less than 0.02 eV Å^{−1}. The neutral molecules CHCl and FAI ion pair was optimized in a 25 Å × 25 Å × 25 Å unit cell. The Eu structures were optimized by the DFT-driven stochastic surface walking method to select the most optimal structures. The 4 × 4 surface models of the (010) and (111) surfaces of FAI were cleaved from the bulk with a 15 Å vacuum layer. An ML–MD simulation was conducted in the canonical ensemble (NVT), with the number of atoms, system volume and temperature kept constant throughout the simulations. The simulation temperature was set to 463 K, in accordance with experimental conditions. The simulation time was 1 ns.

Device characterization

The *J*–*V* measurements of the devices were recorded using a SINUS220 solar simulator (Wavelabs) under 1-sun air mass 1.5 G (100 mW cm^{−2}) illumination in an atmospheric environment, with the temperature controlled to 22 °C. The illumination area of the tandem devices is defined by 1 cm² masks. Operational stability of the unencapsulated devices was measured with MPPT (Shenzhen PURI Materials) under full spectrum light-emitting-diode-simulated 1-sun intensity; these tests were performed in a N₂-filled glovebox. Intensity-dependent photoluminescence measurements were conducted using an LP20-32 radiative efficiency meter (QYB). The laser (532 nm) was switched on 15 min before

the measurement was made to allow the laser to warm up and stabilize. External quantum efficiency measurements were conducted using a Bentham PVE300-IVT system. The light-emitting-diode laser intensity was calibrated with built-in silicon and germanium diodes before measurements. The light was chopped at 137 Hz for the high-resolution EQE measurements and coupled into a Bentham monochromator. The resulting monochromatic light was focused on the devices, and its current under short-circuit conditions was fed to a current preamplifier (Stanford SR 570) before it was analysed with a lock-in amplifier (Stanford SR830 DSP). The time constant of the lock-in amplifier was chosen to be 1 s, and the preamplifier's amplification was increased to resolve low photocurrents. The EQE was determined by dividing the photocurrent of the cell by the flux of incoming photons, which was measured using a calibrated Si photodiode. All EQE measurements were performed under ambient conditions at around 25 °C, with a relative humidity of approximately 70%, and devices without encapsulation.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request.

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Author contributions C.L. and Y.H. conceived the idea and designed the experiments. Y.H., Y.F.C., J.F.G. and X.L.Z. directed and supervised the project. C.L. and R.H. fabricated the tandem devices. Y.S., Y.M. and Z.G.X. assisted with top and bottom cell fabrication. J.C.H. and M.L. performed the TEM measurements. R.L. and W.-J.Y. conducted the theoretical calculation. Y.D.W., X.Y.D. and Q.L.Z. assisted with SEM characterization. C.Y. and S.B. assisted with gas chromatography–mass spectrometry. Z.G. and J.W. performed the in situ photoluminescence test. F.T.P. assisted with the nuclear magnetic resonance tests. Y.D.W. assisted with device fabrication. Y.X.Y., Y.D.W. and C.X.K. assisted with the solar cell test and characterizations. Q.Z., K.L.W., P.G. and Y.B.M. performed the CLSM mapping, TRPL and photoluminescence mapping characterizations. X.X.N. conducted the pseudo *J*–*V* test. L.K.L. performed the tandem device encapsulation. N.X.L., Z.J.D. and X.G. assisted with the operation of the evaporation equipment. Z.H.Z., J.X.C., Z.R.J., X.W., X.M., X.Y.Z. and T.W. assisted with the preparation of other functional layers of the device. Y.C. assisted with the in situ characterization of perovskite films. C.L. and Y.H. wrote the paper. All authors read and commented on the paper.

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Competing interests Y.H. is the founder of Singfilm Solar, a company commercializing perovskite photovoltaics. The other authors declare no competing interests.

Additional information

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

1 cm² and 220 cm²

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Metal aperture mask (black color)

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

Figs. 4c and g

Voltage scan conditions

☒ Yes
☐ No

Scanning rate of 50 mV s⁻¹ (voltage step of 10 mV and delay time of 200 ms) from 2.0 to -0.01 V (reverse) or from -0.01 to 2.0 V (forward).

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

Room temperature and in ambient

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

No preconditioning condition was used.

Stability of the J-V characteristic

☒ Yes
☐ No

Figs. 4e, h

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

Hysteresis is negligible.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

Figs. 4c and g

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

Fig. 4d

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator

☐ Yes
☒ No

State where this information can be found in the text.

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes ☐ No	In the methods section. Explain why this information is not reported/not relevant.
5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	See details in Method. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	See details in Method. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	See details in Method. Explain why this information is not reported/not relevant.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	1 cm ² , 147.5 cm ² and 200 cm ² Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	Report the difference in the short-circuit current density values measured with the mask and aperture area. No significant variations was observed
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Supplementary Figs. 56-58 Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	Supplementary Figs. 56-58 Explain why this information is not reported/not relevant.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	20 Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Fig. 4f Explain why this information is not reported/not relevant.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Storage in dark N ₂ ; MPPT; Damp-heat 85°C/85% RH; out door operation. Explain why this information is not reported/not relevant.